Supporting Information

Dust and Pollen Events Show Similar Aerosol Optical Properties but Distinct Particle Size-Distribution Characteristics in a Finnish Boreal Forest

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S1 Event classification and event-count logic

The main manuscript separates two tasks: event detection and event typing. Event detection compares the pooled event class, $E = D \cup P$, with background observations, B, where D denotes dust-event observations and P denotes pollen-event observations. Event typing compares dust-event observations with pollen-event observations within the event subset.

Dust events were identified using the long-range transported dust-event framework described in the main manuscript and the processed dust-event indicator.[15] Pollen events were identified from SMEAR II impactor-based pollen observations at the SMEAR II station.[7] In the merged optical–PM-mass dataset used for the main event-detection and event-typing analyses, a row was classified as pollen when any of the four processed impactor flag columns equalled the pollen-event code 555. In this processed flag table, the code 555 denotes sampling intervals classified as pollen-affected observations. Background observations were defined as non-event observations, i.e., sampling intervals classified as neither dust nor pollen.

No dust–pollen overlap occurred in the merged optical–PM-mass dataset. Therefore, no priority rule between dust and pollen affected the final event labels.

Table S1: Final exclusive event-class counts in the merged optical–PM-mass dataset. These counts refer to classified sampling intervals and not necessarily to statistically independent event episodes.

Class	Number of observations
Background	1545
Pollen	21
Dust	15
Event	36
Total	1581

Table S2: Event-month distribution in the merged optical–PM-mass dataset.

Event type	Month	Number of observations
Dust	February	2
Dust	March	3
Dust	April	4
Dust	May	1
Dust	June	2
Dust	July	2
Dust	August	1
Pollen	April	2
Pollen	May	14
Pollen	June	5

S2 Optical-only availability versus merged optical–PM-mass analysis counts

Two analysis-ready data structures were used. The optical-only data tables describe instrumental and optical-data availability. The merged optical–PM-mass dataset was used for like-for-like event-detection and event-typing comparisons involving optical variables and PM-mass variables.

The larger optical-only counts should not be interpreted as the sample size used for the merged optical–PM-mass event statistics in the main manuscript. These optical-only counts are provided only to document instrumental data availability; they are not used as the sample sizes for the merged optical–PM-mass event-detection or event-typing effect sizes.

Table S3: Optical-only valid observation counts. These counts describe optical-data availability before restriction to the merged optical–PM-mass event-analysis table.

Variable	Valid observations
PM_1 $\sigma_{sca,550}$	4043
$PM_{1–10}$ $\sigma_{sca,550}$	4019
PM_1 SAE	4043
$PM_{1–10}$ SAE	3981
PM_1 $\sigma_{abs,520}$	3254
$PM_{1–10}$ $\sigma_{abs,520}$	3254
PM_1 AAE	3253
$PM_{1–10}$ AAE	3049
PM_1 SSA_{550}	3198
$PM_{1–10}$ SSA_{550}	3026

Table S4: Variable-specific event counts in the merged optical–PM-mass dataset. Counts are calculated after variable-specific missing-value filtering.

Variable	Valid observations	Background	Dust	Pollen	Event
PM_1 mass	1563	1527	15	21	36
PM_{10} mass	1563	1527	15	21	36
$PM_{1–10}$ mass	1563	1527	15	21	36
Super-PM_{10} mass	1563	1527	15	21	36
PM_1 $\sigma_{sca,550}$	1481	1447	13	21	34

Table S4: Variable-specific event counts in the merged optical–PM-mass dataset. Continued.

Variable	Valid observations	Background	Dust	Pollen	Event
$PM_{1–10}$ $\sigma_{sca,550}$	1474	1440	13	21	34
PM_1 SAE	1481	1447	13	21	34
$PM_{1–10}$ SAE	1469	1436	13	20	33
PM_1 $\sigma_{abs,520}$	1231	1205	11	15	26
$PM_{1–10}$ $\sigma_{abs,520}$	1231	1205	11	15	26
PM_1 AAE	1231	1205	11	15	26
$PM_{1–10}$ AAE	1205	1180	11	14	25
PM_1 SSA_{550}	1214	1191	9	14	23
$PM_{1–10}$ SSA_{550}	1193	1170	9	14	23
PM_1 MAC_{520}	1202	1176	11	15	26
$PM_{1–10}$ MAC_{520}	1202	1176	11	15	26
PM_1 MSC_{550}	1451	1417	13	21	34
$PM_{1–10}$ MSC_{550}	1441	1407	13	21	34

S3 Measurement site, instruments, and derived variables

S3.1 Measurement site

The measurements used in this study were conducted at the Station for Measuring Ecosystem–Atmosphere Relations II (SMEAR II), Hyytiälä, southern Finland. SMEAR II is a long-term boreal forest research station designed for continuous measurements of ecosystem–atmosphere exchange, trace gases, meteorology, and atmospheric aerosol properties.[7] The station has been widely used for long-term aerosol number size-distribution, aerosol optical-property, and new-particle-formation studies in a boreal environment.[4, 3, 10, 1]

S3.2 Aerosol optical-property instruments

Aerosol optical properties were derived from in-situ measurements of particle light scattering and light absorption. Long-term aerosol optical measurements at SMEAR II, including size-resolved PM_1, PM_{10}, and supermicron aerosol optical properties, have been described previously.[10, 1] In the present analysis, the main optical variables included the scattering coefficient at 550 nm, $\sigma_{sca,550}$; the absorption coefficient at 520 nm, $\sigma_{abs,520}$; the scattering Ångström exponent (SAE); the absorption Ångström exponent (AAE); the single-scattering albedo at 550 nm, SSA_{550}; the mass absorption coefficient, MAC_{520}; and the mass scattering coefficient, MSC_{550}. Optical variables were analysed separately for the PM_1 and $PM_{1–10}$ size ranges where available. These variables were used to test whether dust and pollen events produce distinguishable optical responses or whether the optical signatures overlap.

S3.3 PM-mass variables

Size-resolved PM-mass variables were taken from the merged optical–PM-mass analysis table. The PM-mass variables used in the present study were PM_1 mass, PM_{10} mass, $PM_{1–10}$ mass, and super-PM_{10} mass. Here, $PM_{1–10}$ mass denotes the derived mass difference between PM_{10} and PM_1, and super-PM_{10} mass denotes the derived mass fraction above 10 μm. Because some PM-mass quantities are subtraction-derived, negative values can occur when the relevant measured or derived quantities are close to their uncertainty limits. The negative-value audit and sensitivity analysis are reported in Sect. S6.

S3.4 Particle size-distribution instruments and APSD-derived metrics

Aerosol particle size distributions were constructed from mobility and aerodynamic particle size-distribution measurements. The fine-particle size range was measured using a differential mobility particle sizer framework, while the larger-particle size range was measured using an aerodynamic particle sizer. This combined DMPS–APS framework is commonly used to describe the particle number size distribution across submicron and supermicron sizes at SMEAR II and related long-term atmospheric aerosol stations.[10, 17, 1] In the present analysis, the merged size distributions were expressed as number size distributions, cross-sectional-area size distributions, and volume size distributions.

The APSD-derived metrics were calculated from these size distributions to connect the measured particle-size structure with optical and mass responses. The main derived metrics were accumulation-mode cross-sectional area, S_{acc}; accumulation-mode volume, V_{acc}; coarse-mode volume, V_{coarse}; the volume-ratio metric, R_V; and Aitken-mode number concentration, N_{Ait}. The number distribution emphasizes particle abundance, the cross-sectional-area distribution emphasizes scattering-relevant particle geometry, and the volume distribution provides a closer link to particle mass for particles of comparable density and morphology.[13] Because mobility and aerodynamic instruments characterize particle size using different equivalent diameters, the APSD-derived area and volume metrics should be interpreted as operational size-distribution metrics unless aerodynamic diameters are converted to geometric or volume-equivalent diameters using explicit density and shape assumptions.

The optical, PM-mass, and APSD-derived datasets differ in time resolution, data availability, and filtering history. Therefore, the main manuscript distinguishes between optical-only availability, merged optical–PM-mass event-analysis counts, and native-resolution APSD-derived metric counts. The optical-only counts describe instrumental data availability, whereas the merged optical–PM-mass counts are used for like-for-like event-detection and event-typing comparisons involving optical and PM-mass variables. The APSD-derived metrics are analysed at native time resolution and are therefore interpreted together with temporal-dependence sensitivity checks.

S4 Statistical definitions and unit of inference

For each candidate variable X, event detection was defined as the comparison between the pooled event class and background:

$$\Delta^*_{\text{det}}(X) = \frac{\text{median}(X|E) - \text{median}(X|B)}{\text{IQR}(X|B)}. \tag{S1}$$

Dust–pollen typing was defined as:

$$\Delta^*_{\text{typ}}(X) = \frac{\text{median}(X|D) - \text{median}(X|P)}{\text{IQR}(X|B)}. \tag{S2}$$

Here, E is the pooled event class, B is background, D is dust, and P is pollen. Positive typing values indicate larger values during dust events, whereas negative typing values indicate larger values during pollen events.

Cliff's δ was used as a non-parametric effect-size measure:[2, 16]

$$\delta = \frac{\#(x_i > y_j) - \#(x_i < y_j)}{n_x n_y}. \tag{S3}$$

Bootstrap confidence intervals were calculated for the median difference between groups.[6] Mann–Whitney tests were used as non-parametric distributional tests.[11] For the native-resolution APSD-derived metrics, p values are treated as supporting diagnostics because repeated observations within event periods are temporally autocorrelated. The main APSD interpretation is therefore based on effect direction, standardized effect size, confidence intervals, persistence after month and month-plus-hour background correction, and event-declustered sensitivity checks where available.

Because the number of independent dust and pollen events was small, especially after event declustering, the results should be interpreted as evidence for physically interpretable event-type tendencies rather than as a definitive operational classification model.

S5 Merged optical–PM-mass effect-size results

S5.1 Event detection

Table S5: Event-detection effect sizes in the merged optical–PM-mass dataset. The contrast is event minus background. PM mass is reported in μg m^{-3}; scattering and absorption coefficients are reported in Mm^{-1}; SAE, AAE, and SSA are dimensionless; MAC and MSC are reported in their corresponding mass-normalized coefficient units.

Variable	Δ^*_{det}	Median difference	CI low	CI high	Cliff's δ
PM_1 mass	1.014	2.646	1.555	3.875	0.489
PM_{10} mass	1.414	5.507	3.301	9.567	0.581
$PM_{1\text{–}10}$ mass	1.642	2.438	1.283	7.069	0.619
Super-PM_{10} mass	3.721	2.918	1.558	5.677	0.712
PM_1 $\sigma_{\mathrm{sca},550}$	0.883	8.046	3.847	12.287	0.463
$PM_{1\text{–}10}$ $\sigma_{\mathrm{sca},550}$	0.806	1.823	0.478	2.885	0.342
PM_1 SAE	0.378	0.156	0.079	0.213	0.284
$PM_{1\text{–}10}$ SAE	0.159	0.079	-0.139	0.273	0.060
PM_1 $\sigma_{\mathrm{abs},520}$	0.452	0.552	0.006	1.001	0.277
$PM_{1\text{–}10}$ $\sigma_{\mathrm{abs},520}$	0.706	0.086	-0.005	0.137	0.248
PM_1 AAE	-0.224	-0.042	-0.080	0.022	-0.176
$PM_{1\text{–}10}$ AAE	0.000	0.000	-0.067	0.161	0.144
PM_1 SSA_{550}	0.723	0.056	-0.004	0.073	0.311
$PM_{1\text{–}10}$ SSA_{550}	-0.098	-0.003	-0.012	0.012	-0.028
PM_1 MAC_{520}	-0.483	-0.208	-0.300	-0.041	-0.293
$PM_{1\text{–}10}$ MAC_{520}	-0.560	-0.049	-0.059	-0.025	-0.440
PM_1 MSC_{550}	-0.192	-0.287	-0.632	0.365	-0.079
$PM_{1\text{–}10}$ MSC_{550}	-0.444	-0.660	-1.152	-0.348	-0.380

S5.2 Dust–pollen typing

Table S6: Dust–pollen typing effect sizes in the merged optical–PM-mass dataset. The contrast is dust minus pollen. Negative values indicate larger values during pollen events. PM mass is reported in μg m^{-3}; scattering and absorption coefficients are reported in Mm^{-1}; SAE, AAE, and SSA are dimensionless; MAC and MSC are reported in their corresponding mass-normalized coefficient units.

Variable	Δ^*_{typ}	Median difference	CI low	CI high	Cliff's δ
PM_1 mass	0.408	1.064	-3.593	2.557	-0.060
PM_{10} mass	-0.640	-2.493	-10.263	2.508	-0.283
$PM_{1\text{–}10}$ mass	-3.253	-4.830	-7.264	-0.527	-0.435
Super-PM_{10} mass	-4.932	-3.867	-9.032	0.105	-0.594
PM_1 $\sigma_{\mathrm{sca},550}$	0.689	6.273	-4.100	12.788	0.275
$PM_{1\text{–}10}$ $\sigma_{\mathrm{sca},550}$	0.461	1.044	-1.690	2.876	0.084
PM_1 SAE	-0.071	-0.029	-0.296	0.198	0.048
$PM_{1\text{–}10}$ SAE	-0.628	-0.311	-0.646	0.179	-0.338
PM_1 $\sigma_{\mathrm{abs},520}$	0.082	0.100	-1.083	0.798	-0.018
$PM_{1\text{–}10}$ $\sigma_{\mathrm{abs},520}$	-0.791	-0.097	-0.191	0.093	-0.248
PM_1 AAE	0.421	0.078	-0.119	0.103	0.079
$PM_{1\text{–}10}$ AAE	0.561	0.184	-0.065	0.445	0.377
PM_1 SSA_{550}	0.352	0.027	-0.094	0.095	0.063
$PM_{1\text{–}10}$ SSA_{550}	-0.410	-0.013	-0.071	0.013	-0.349

Table S6: Dust–pollen typing effect sizes in the merged optical–PM-mass dataset. Continued.

Variable	Δ^*_{typ}	Median difference	CI low	CI high	Cliff's δ
$\mathrm{PM_1}$ $\mathrm{MAC_{520}}$	0.012	0.005	-0.189	0.711	0.224
$\mathrm{PM_{1\text{–}10}}$ $\mathrm{MAC_{520}}$	0.195	0.017	-0.016	0.113	0.273
$\mathrm{PM_1}$ $\mathrm{MSC_{550}}$	0.869	1.303	-0.173	2.319	0.502
$\mathrm{PM_{1\text{–}10}}$ $\mathrm{MSC_{550}}$	0.407	0.605	-0.213	1.795	0.407

S6 Negative derived mass-value audit

Negative values occurred only under background conditions for the PM-mass variables listed in Table S7. These negative values are expected to be most relevant for subtraction-derived quantities, especially super-$\mathrm{PM_{10}}$ mass. Sensitivity tests excluding negative mass values produced only minor changes in the main effect-size estimates.

Table S7: Negative derived mass-value audit in the merged optical–PM-mass dataset. PM mass is reported in μg m^{-3}.

Variable	Finite values	Negative values	Negative fraction	Negative background	Negative event
$\mathrm{PM_1}$ mass	1563	2	0.0013	2	0
$\mathrm{PM_{10}}$ mass	1563	5	0.0032	5	0
$\mathrm{PM_{1\text{–}10}}$ mass	1563	28	0.0179	28	0
Super-$\mathrm{PM_{10}}$ mass	1563	69	0.0441	69	0

Table S8: Sensitivity of selected standardized effect sizes to excluding negative mass values.

Variable	Detection retained	Detection excl. negative	Change	Typing retained	Typing excl. negative	Change
$\mathrm{PM_1}$ mass	1.014	1.015	0.001	0.408	0.408	0.000
$\mathrm{PM_{10}}$ mass	1.414	1.419	0.005	-0.640	-0.643	-0.003
$\mathrm{PM_{1\text{–}10}}$ mass	1.642	1.620	-0.022	-3.253	-3.244	0.009
Super-$\mathrm{PM_{10}}$ mass	3.721	3.584	-0.137	-4.932	-4.811	0.121

S7 APSD-derived metric results

The APSD-derived metrics were analysed at native time resolution. Therefore, their timestamp-level sample sizes are much larger than the merged optical–PM-mass dataset. These measurements are repeated, temporally autocorrelated observations within event periods. The native-resolution results are therefore interpreted together with event-declustered and moving-block-bootstrap sensitivity checks, where available.

The APSD metric units are as follows: S_{acc} is reported in μm^2 cm^{-3}, V_{acc} and V_{coarse} are reported in μm^3 cm^{-3}, R_V is dimensionless, and N_{Ait} is reported in cm^{-3}.

Table S9: Native-resolution APSD-derived metric counts and medians.

Metric	Class	Non-missing values	Median	IQR	Unit
S_{acc}	Background	111014	30.222	42.091	μm^2 cm^{-3}
S_{acc}	Dust	923	71.865	89.542	μm^2 cm^{-3}
S_{acc}	Pollen	1381	49.123	58.809	μm^2 cm^{-3}
V_{acc}	Background	111014	1.199	1.749	μm^3 cm^{-3}
V_{acc}	Dust	923	2.808	3.464	μm^3 cm^{-3}
V_{acc}	Pollen	1381	1.864	2.304	μm^3 cm^{-3}
V_{coarse}	Background	111648	1.204	1.432	μm^3 cm^{-3}

Table S9: Native-resolution APSD-derived metric counts and medians. Continued.

Metric	Class	Non-missing values	Median	IQR	Unit
V_{coarse}	Dust	923	1.369	1.489	μm^3 cm^{-3}
V_{coarse}	Pollen	1381	1.383	2.085	μm^3 cm^{-3}
R_V	Background	111014	0.945	1.736	dimensionless
R_V	Dust	923	0.518	1.100	dimensionless
R_V	Pollen	1381	0.714	1.494	dimensionless
N_{Ait}	Background	111364	404.605	525.627	cm^{-3}
N_{Ait}	Dust	923	412.412	455.336	cm^{-3}
N_{Ait}	Pollen	1381	693.805	766.068	cm^{-3}

Table S10: Pool B month-plus-hour dust–pollen anomaly contrasts for APSD-derived metrics at native timestamp resolution. The contrast is dust minus pollen. Median differences and confidence intervals are reported in the unit of each metric.

Metric	Dust n	Pollen n	Median difference	CI low	CI high	Cliff's δ	Δ^*	Unit
S_{acc}	923	1381	25.390	17.062	31.983	0.213	0.558	μm^2 cm^{-3}
V_{acc}	923	1381	1.000	0.686	1.238	0.196	0.520	μm^3 cm^{-3}
V_{coarse}	923	1381	-0.057	-0.169	0.079	-0.018	-0.036	μm^3 cm^{-3}
R_V	923	1381	-0.118	-0.190	-0.040	-0.135	-0.064	dimensionless
N_{Ait}	923	1381	-131.553	-177.840	-87.879	-0.137	-0.250	cm^{-3}

For S_{acc}, the native-resolution analysis showed a persistent positive dust–pollen contrast under raw, common-year–month, month-anomaly, and month-plus-hour-anomaly comparisons. Under the preferred Pool B month-plus-hour anomaly baseline, the dust-minus-pollen median difference was 25.390 μm^2 cm^{-3}, with a confidence interval of 17.062–31.983 μm^2 cm^{-3} and $\Delta^* = 0.558$. After event declustering, the contrast retained the same positive direction and had a larger standardized effect, but the bootstrap confidence interval crossed zero. Therefore, S_{acc} is interpreted as the strongest APSD-based contrast in this dataset, rather than as a fully validated source classifier.

For V_{acc}, the raw and native-resolution anomaly analyses indicated higher accumulation-mode volume during dust events than during pollen events. Under the preferred Pool B month-plus-hour native-resolution anomaly baseline, the dust-minus-pollen median difference was 1.000 μm^3 cm^{-3} with $\Delta^* = 0.520$. The timestamp-level moving-block bootstrap also supported a positive contrast, with a median difference of 1.033 μm^3 cm^{-3} and a confidence interval of 0.026–1.969 μm^3 cm^{-3}. However, after event declustering into independent event medians, the dust-minus-pollen median difference was 1.390 μm^3 cm^{-3} with $\Delta^* = 0.723$, but the confidence interval crossed zero, from -0.453 to 3.294 μm^3 cm^{-3}. Thus, V_{acc} supports the accumulation-mode interpretation but is less robust than S_{acc} as a dust–pollen discriminator.

For V_{coarse}, the event-declustered month-plus-hour anomaly analysis gave the same qualitative conclusion as the native-resolution analysis: the dust–pollen confidence interval crossed zero. The event-level dust-minus-pollen median difference was -0.698 μm^3 cm^{-3}, with a confidence interval of -2.180 to 0.903 μm^3 cm^{-3} and $\Delta^* = -0.442$, while the timestamp-level moving-block bootstrap also crossed zero. Therefore, V_{coarse} is not interpreted as a robust dust–pollen discriminator.

For R_V, the native-resolution analysis indicated a small negative dust–pollen contrast. However, after event declustering into independent event medians, the Pool B month-plus-hour anomaly contrast was not robust: the dust-minus-pollen median difference was -0.439, with a confidence interval of -1.660 to 0.611 and $\Delta^* = -0.238$. The timestamp-level moving-block bootstrap also crossed zero. Therefore, R_V is interpreted only as weak supporting context and not as a primary dust–pollen discriminator.

Table S11: Event-declustered and moving-block-bootstrap sensitivity checks for APSD-derived metrics where available. Event-declustered results use independent event medians after month-plus-hour anomaly correction. Moving-block results use timestamp-level month-plus-hour anomalies with 24 h blocks. The contrast is dust minus pollen. Median differences and confidence intervals are reported in the unit of each metric.

Metric	Event diff.	Event CI low	Event CI high	Event Δ^*	Block diff.	Block CI low	Block CI high	Unit
S_{acc}	41.390	-7.028	64.279	0.910	–	–	–	$\mu\mathrm{m}^2\ \mathrm{cm}^{-3}$
V_{acc}	1.390	-0.453	3.294	0.723	1.033	0.026	1.969	$\mu\mathrm{m}^3\ \mathrm{cm}^{-3}$
V_{coarse}	-0.698	-2.180	0.903	-0.442	-0.051	-0.674	0.494	$\mu\mathrm{m}^3\ \mathrm{cm}^{-3}$
R_V	-0.439	-1.660	0.611	-0.238	-0.136	-0.593	0.201	dimensionless
N_{Ait}	-231.405	-477.830	56.167	-0.439	-126.517	-344.962	28.537	cm^{-3}

Overall, S_{acc} is treated as the strongest APSD-based dust–pollen contrast in this dataset. V_{acc} provides supporting accumulation-mode evidence in the same direction but is less robust after event declustering. V_{coarse} and R_V are not treated as robust primary discriminators. N_{Ait} is treated as process context rather than as a primary dust–pollen typing metric.

S8 Reconciling PM mass and accumulation-mode cross-sectional area

The main manuscript reports that $\mathrm{PM}_{1\text{–}10}$ mass was higher during pollen events than during dust events in the merged optical–PM-mass event set, whereas S_{acc} was higher during dust events after month-plus-hour background correction. These results are not contradictory because PM mass and cross-sectional area weight the particle size distribution differently.

PM mass is approximately volume-weighted and scales with particle diameter cubed.[13] Therefore, $\mathrm{PM}_{1\text{–}10}$ mass can be strongly influenced by relatively large supermicron and coarse-tail particles. In contrast, S_{acc} is an accumulation-mode cross-sectional-area metric and scales with particle diameter squared. It is therefore more directly linked to scattering-relevant particle geometry in the accumulation-mode size range. This distinction explains why pollen can show higher $\mathrm{PM}_{1\text{–}10}$ mass in the present event set, while dust can show higher controlled accumulation-mode cross-sectional area.

S9 Sub-100 nm and NPF-related diagnostics: NPF as a process not specific to dust or pollen

The sub-100 nm size range was analysed to determine whether dust and pollen event periods differed in their nucleation- and Aitken-mode particle behaviour. The 100 nm diameter was used as a diagnostic division because it approximately separates the nucleation- and Aitken-mode size range from the larger accumulation- and coarse-mode size range. This division is not interpreted as a strict physical threshold. Rather, it provides a practical way to separate particle sizes most directly associated with new-particle formation (NPF), early growth, and Aitken-mode variability from the larger size range that contributes more strongly to aerosol cross-sectional area, volume, PM mass, scattering, and cloud condensation nuclei (CCN) relevance.

The central interpretation from the sub-100 nm diagnostics is that NPF-like behaviour is not specific to either dust events or pollen events. In this context, this means that NPF-like particle evolution should not be interpreted as a source-specific signature of dust or pollen. Instead, it reflects the balance between process-level drivers that can occur during either event type. These drivers include the production of low-volatility vapours, photochemical activity, boundary-layer evolution, particle growth rate, and losses to the pre-existing aerosol population through condensation and coagulation sinks.[8, 4, 3, 14] Therefore, an event labelled as pollen or dust can coincide with NPF-like behaviour if the local atmospheric conditions favour nucleation and early particle growth, but either event type can also occur without clear NPF-like behaviour if growth is weak or if the pre-existing aerosol sink is large.

This distinction is important for source interpretation. Dust and pollen are event classes defined by source-related or observational criteria, whereas NPF is a microphysical process. Dust events may increase the pre-existing aerosol surface area and thereby strengthen the condensation or coagulation sink, which can reduce the survival probability of newly formed particles. However, a dust event can also coincide with meteorological or photochemical conditions favourable for NPF. Similarly, pollen periods can coincide with spring or early-summer conditions favourable for biogenic vapour production and photochemistry, but pollen presence alone does not imply that NPF must occur. Thus, sub-100 nm variability should not be interpreted as a direct source fingerprint of pollen or dust. It is better interpreted as process-level context describing whether nucleation-mode and Aitken-mode particle dynamics were active during the event period.

Raw N_{Ait} was higher during pollen events than during dust events, with median values of 693.805 cm^{-3} for pollen, 412.412 cm^{-3} for dust, and 404.605 cm^{-3} for background. However, after month-plus-hour anomaly correction and event declustering, the difference between dust and pollen retained the same direction but was not statistically robust: the event-level dust-minus-pollen median difference was -231.405 cm^{-3}, with a confidence interval of -477.830 to 56.167 cm^{-3} and $\Delta^* = -0.439$. A timestamp-level moving-block bootstrap gave a similar non-robust result, with a median difference of -126.517 cm^{-3} and a confidence interval of -344.962 to 28.537 cm^{-3}. Thus, N_{Ait} supports the interpretation that pollen periods can be associated with enhanced sub-100 nm particle concentrations, but this contrast is not sufficiently robust after temporal control to be used as a criterion for distinguishing dust from pollen.

Table S12: Summary of N_{Ait} results used to interpret sub-100 nm particle behaviour. The raw difference between pollen and dust provides process context, but the controlled and event-declustered results show that N_{Ait} is not a robust metric for distinguishing dust from pollen.

Quantity	Class or contrast	Value
Raw median	Background	404.605 cm^{-3}
Raw median	Dust	412.412 cm^{-3}
Raw median	Pollen	693.805 cm^{-3}
Event-declustered median difference	Dust–pollen	-231.405 cm^{-3}
Event-declustered CI	Dust–pollen	-477.830 to 56.167 cm^{-3}
Event-declustered Δ^*	Dust–pollen	-0.439
Moving-block median difference	Dust–pollen	-126.517 cm^{-3}
Moving-block CI	Dust–pollen	-344.962 to 28.537 cm^{-3}

The event-resolved sub-100 nm diagnostics are shown in Figures S1–S6. These figures show particle number size distributions (PSD), volume size distributions (VSD), and cross-sectional-area size distributions (ASD) separately for pollen and dust events. The Supporting Information figure files are `figure_5.png`, `figure_6.png`, `figure_7.png`, `figure_8.png`, `figure_9.png`, and `figure_10.png`. Together, these figures show that sub-100 nm particle behaviour varies among individual events and that NPF-like behaviour can occur during both pollen and dust events. Therefore, sub-100 nm behaviour is interpreted as process-level context rather than as a robust stand-alone criterion for distinguishing dust from pollen.

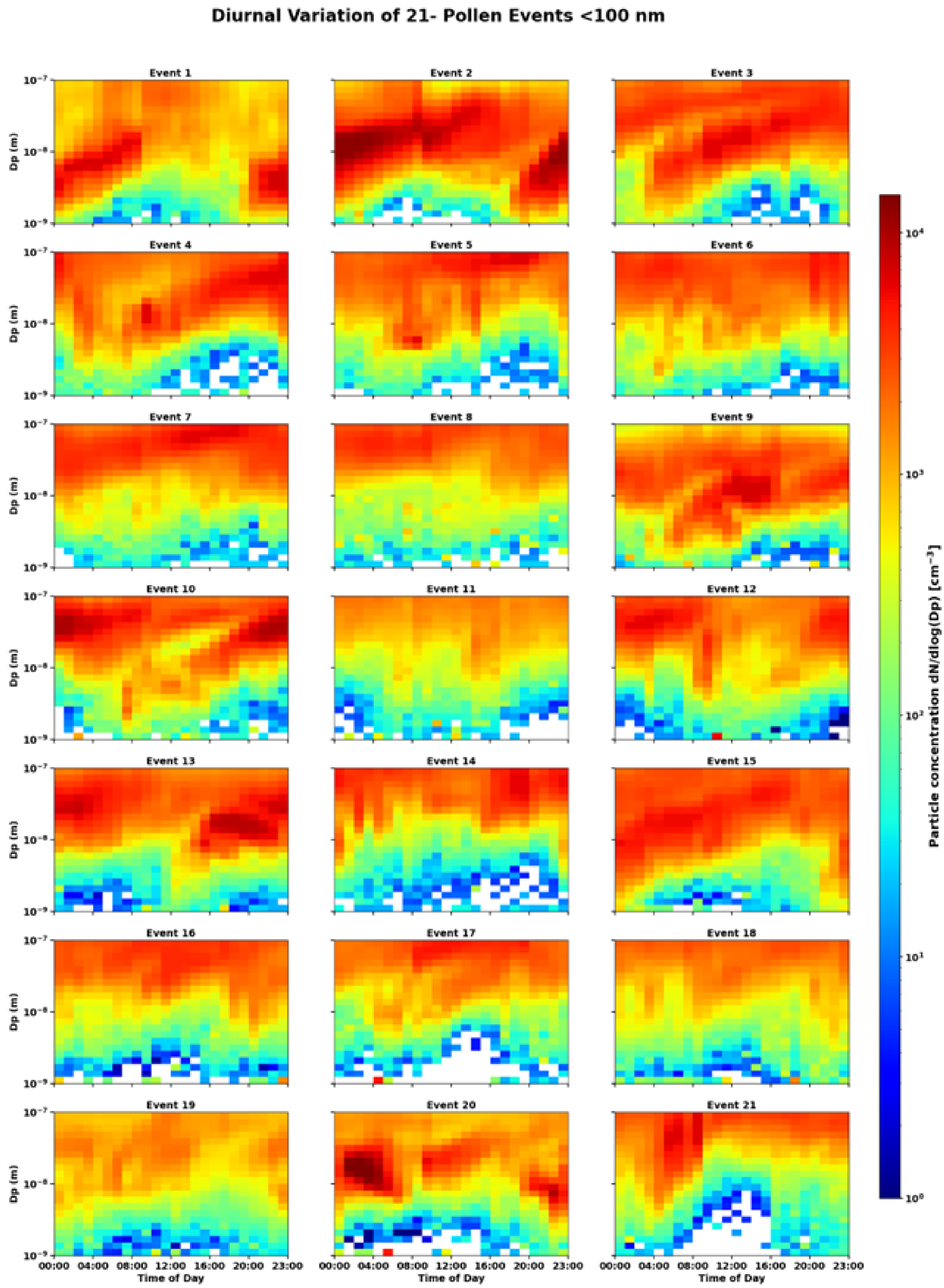

Figure S1: Diurnal variation of sub-100 nm particle number size distributions during the 21 pollen events. The colour scale shows PSD, $\mathrm{d}N/\mathrm{d}\log(D_p)$, in cm^{-3}. Individual panels show the diurnal evolution for each pollen event. These event-resolved PSD panels indicate that pollen-event periods can be associated with enhanced sub-100 nm particle concentrations, but the timing and structure vary among events. Therefore, this figure is interpreted as process-level NPF context rather than as evidence for a pollen-specific NPF signature.

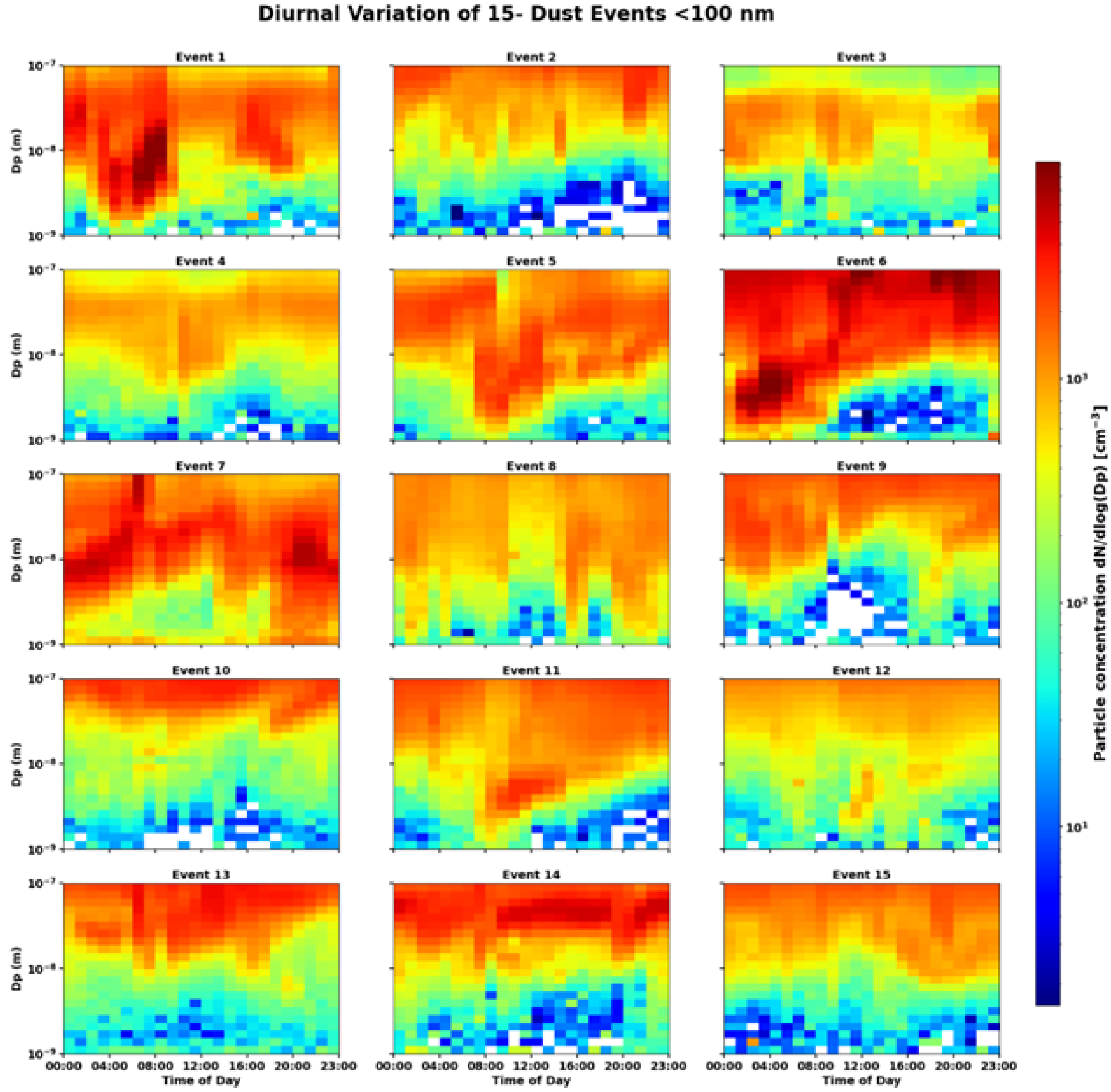

Figure S2: Diurnal variation of sub-100 nm particle number size distributions during the 15 dust events. The colour scale shows PSD, $dN/d\log(D_p)$, in cm^{-3}. Individual panels show the diurnal evolution for each dust event. These dust-event PSD panels show that sub-100 nm particle structure is also present during some dust events, supporting the interpretation that NPF-like behaviour is not unique to pollen periods.

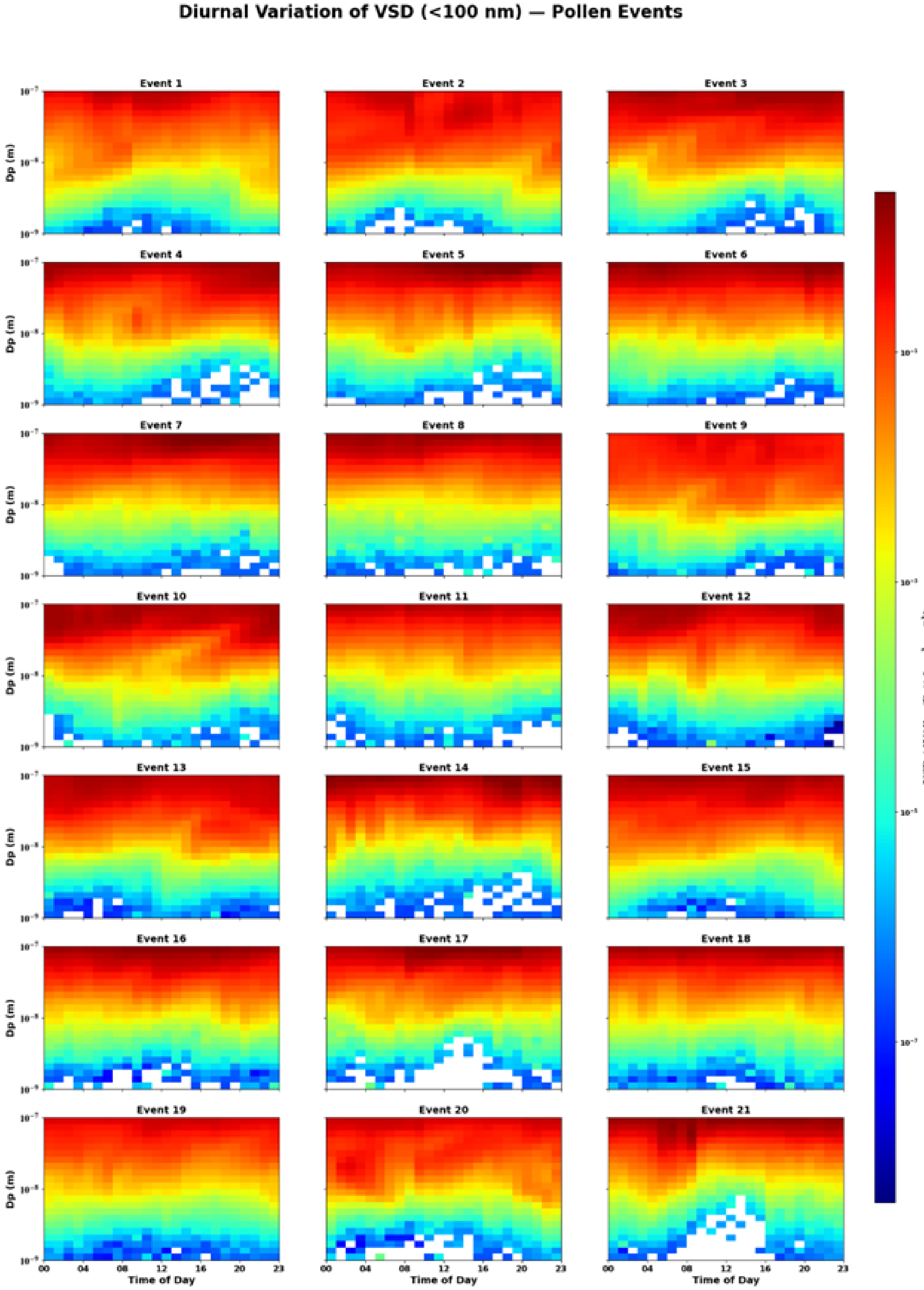

Figure S3: Diurnal variation of sub-100 nm volume size distributions during the 21 pollen events. The colour scale shows VSD, $dV/d\log(D_p)$, in μm^3 cm^{-3}. Individual panels show the diurnal evolution for each pollen event. These panels provide volume-weighted context for the pollen-event sub-100 nm particle population.

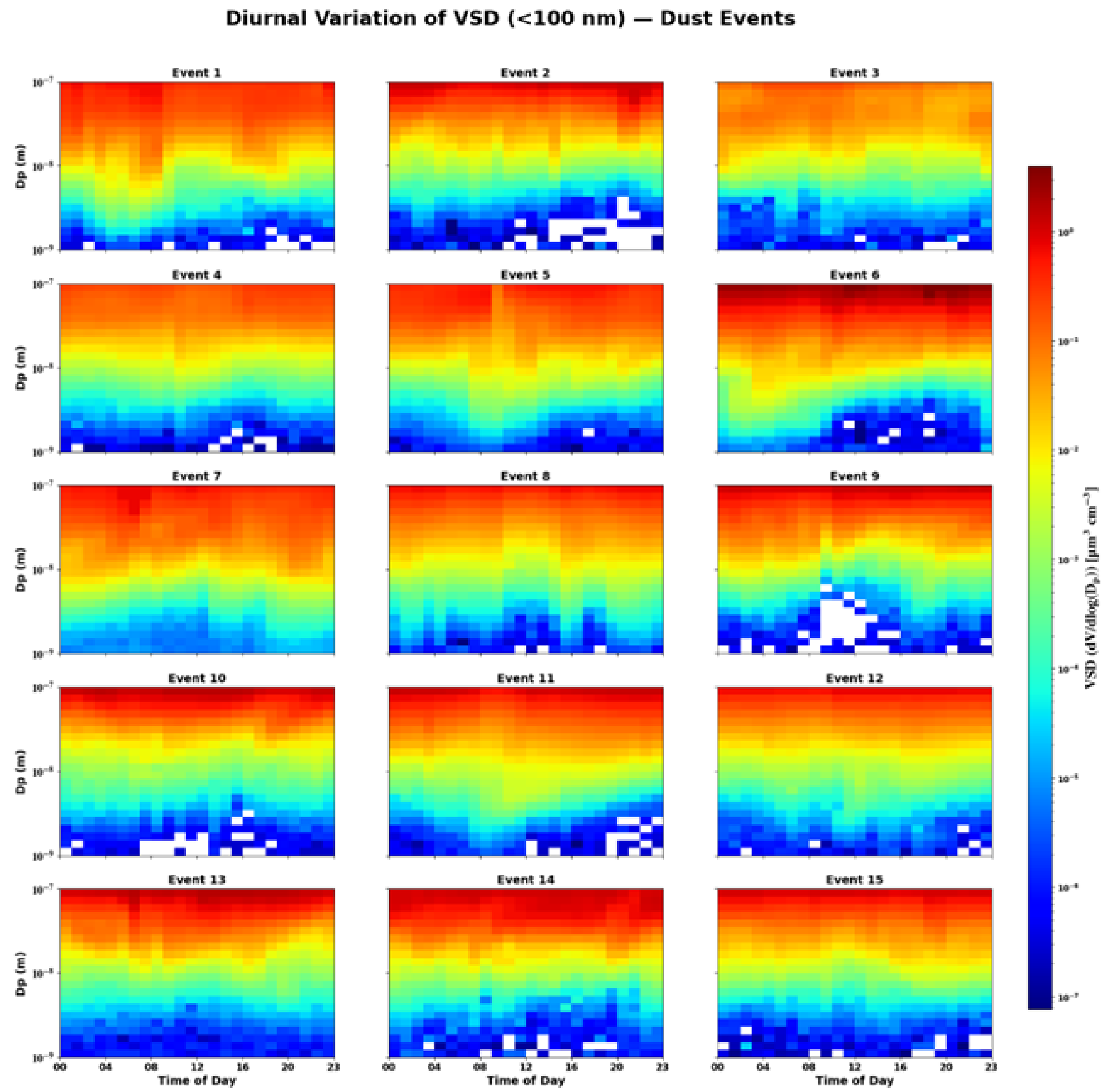

Figure S4: Diurnal variation of sub-100 nm volume size distributions during the 15 dust events. The colour scale shows VSD, $\mathrm{d}V/\mathrm{d}\log(D_p)$, in μm^3 cm^{-3}. Individual panels show the diurnal evolution for each dust event. These panels provide volume-weighted context for the dust-event sub-100 nm particle population.

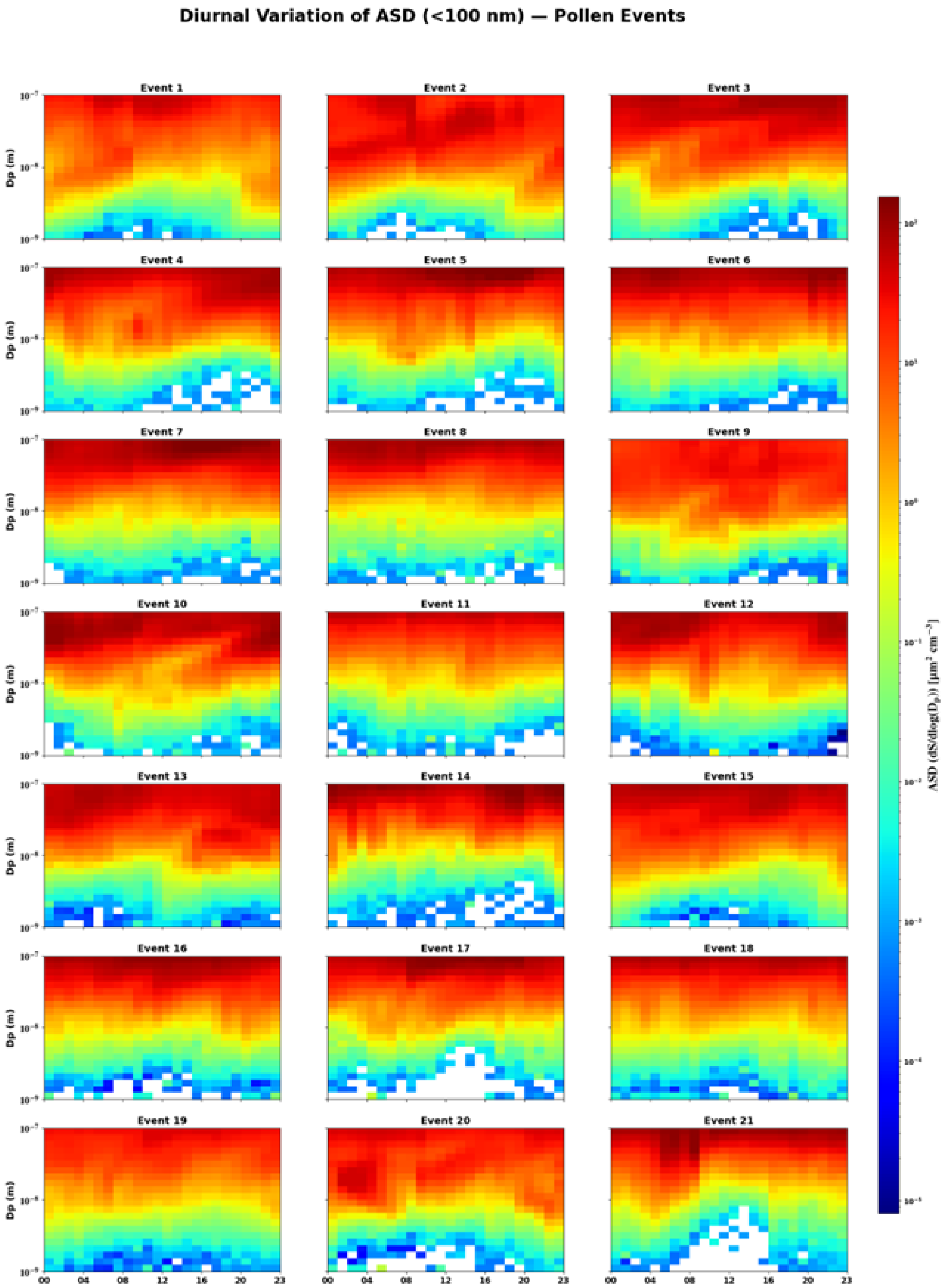

Figure S5: Diurnal variation of sub-100 nm cross-sectional-area size distributions during the 21 pollen events. The colour scale shows ASD, $\mathrm{d}S/\mathrm{d}\log(D_p)$, in $\mu\mathrm{m}^2$ cm^{-3}. Individual panels show the diurnal evolution for each pollen event. These panels provide area-weighted context for the pollen-event sub-100 nm particle population.

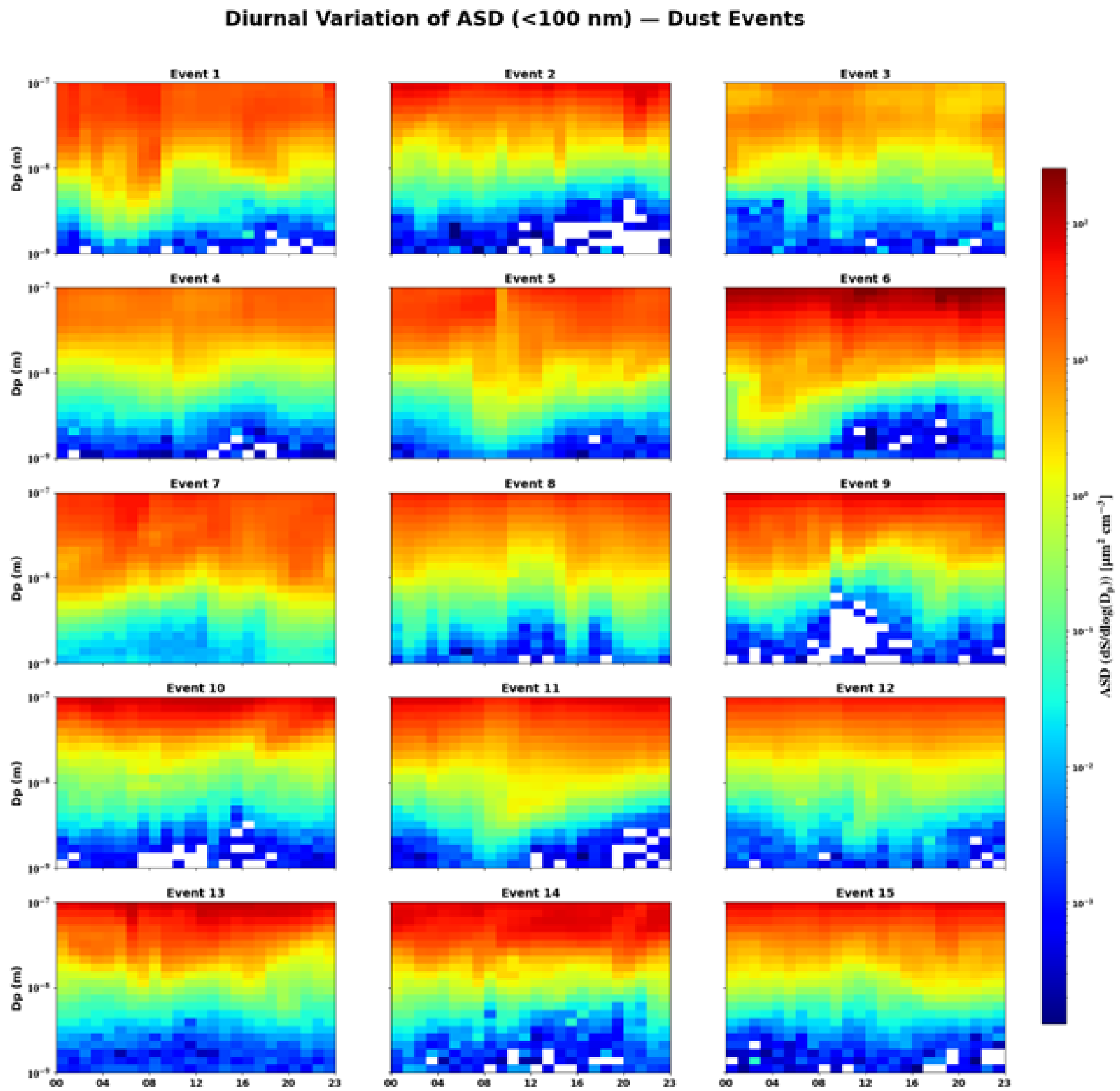

Figure S6: Diurnal variation of sub-100 nm cross-sectional-area size distributions during the 15 dust events. The colour scale shows ASD, $\mathrm{d}S/\mathrm{d}\log(D_p)$, in $\mu\mathrm{m}^2\ \mathrm{cm}^{-3}$. Individual panels show the diurnal evolution for each dust event. These panels provide area-weighted context for the dust-event sub-100 nm particle population.

The process-level interpretation is therefore as follows. Pollen and dust events can modify the aerosol population, but NPF-like sub-100 nm behaviour is controlled by the local balance between precursor vapour supply, particle growth, and removal to the pre-existing aerosol population. Because these controlling factors can occur during either dust or pollen periods, NPF is not treated here as a dust- or pollen-specific source marker. Instead, it is treated as an accompanying aerosol microphysical process that helps contextualize the event environment. CCN relevance is likewise not determined by the event label alone, because CCN activation depends on particle size, hygroscopicity, composition, mixing state, and supersaturation.[5, 12, 9]

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